Sharp denominator corrections for Brown–Zudilin cellular linear forms in $\zeta(5)$: measurements, a conjecture, and a lattice mechanism

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Abstract

Brown and Zudilin [arXiv:2210.03391] constructed effective rational approximations $I'(\mathbf{a}\cdot n) = Q_n\zeta(5) - P_n$ to $\zeta(5)$ from cellular integrals on $\overline{\mathcal{M}}_{0,8}$, together with experimental integrality statements for the coefficients. By recovering the exact rational coefficients $P_n, \hat{P}_n$ at more than 200 parameter points via certified integer-relation computations, we show that the published integrality claims are false as stated—their own printed values violate them—and that the correct statements carry a correction factor of exactly $12 = 2^2 \cdot 3$ (respectively 2 for the $\zeta(3)$-companion), which we conjecture to be sharp and verify in every measured cell. Unlike the familiar correction factors in the classical literature (Apéry, Rhin–Viola, Krattenthaler–Rivoal), which we check are uniformly *removable*, this correction is *attained*. We propose a mechanism: the coefficients are periods against a non-primitive integral Betti class produced by the construction's motivic elimination of $\zeta(2)$, so the correction is a lattice index bounded by the Bernoulli constant 24 arising from $\zeta(2) = -(2\pi i)^2/24$; an elementary elimination-cost lemma, a conditional theorem, and a concrete derivation in the totally symmetric case (where the factor 12 appears as a doubled sine-kernel Laurent coefficient $\frac{1}{6}$) support this. As by-products we determine the supremum of the Brown–Zudilin "worthiness" exponent over the full parameter cone, verify that the leading factor 2 in a denominator conjecture of Krattenthaler–Rivoal is unnecessary for $n \leq 5$, and document an orbit-coherent boundary-prime anomaly. All measured denominator slack is bounded; no measured direction offers exponential arithmetic gain.

1 Introduction

For the linear forms of Apéry-type used in irrationality proofs, the arithmetic of the rational coefficients is traditionally expressed by *inclusions*: statements of the shape $D_n \cdot (\text{coefficient}) \in \mathbb{Z}$ for explicit denominators D_n built from $d_n = \operatorname{lcm}(1, \ldots, n)$. These inclusions are proved (or conjectured) from term-by-term analysis, and the question of their *sharpness*—whether the true denominators actually attain D_n—is rarely addressed and, to our knowledge, has never been investigated systematically by direct computation.

This paper does so for the family of cellular integrals $I(\mathbf{a})$ on $\overline{\mathcal{M}}_{0,8}$ introduced by Brown and Zudilin [1], which produce simultaneous rational approximations $I' = Q\zeta(5) - P$, $I'' = Q\zeta(3) - \hat{P}$ by a motivic elimination of the parasitic $\zeta(2)$. Our findings:

*Draft of 16 July 2026. Computations, conjectures and proofs in this paper were produced in a human–AI collaboration; the verification code and complete datasets are available in the accompanying repository. Correspondence: [address].

1. The integrality statements printed in [1] (their experimental inclusions for I', and the totally symmetric display $d_n^5 P_n \in \mathbb{Z}$) are **false as stated**, already against the paper's own printed values (§3).

2. The corrected statements carry a factor of exactly 12 (resp. 2), and this factor is **sharp**: it is attained in 46 of 164 systematically sampled cells at the prime 2 and 30 cells at the prime 3, and never exceeded in any of our more than 200 certified cells (§4). One isolated, orbit-coherent, non-persistent exception at a boundary prime $p = 7$ is documented in §5.

3. By contrast, the analogous correction factors appearing in the classical literature are uniformly *removable*: we verify $d_n^3 a_n \in \mathbb{Z}$ for Apéry's $\zeta(3)$ forms for $n \leq 50$ (sharpening the ubiquitous "$2d_n^3$"), and we verify that the leading factor 2 retained even in the *conjectured* sharp denominator law of Krattenthaler–Rivoal [3, eq. (17.1)] for Zudilin's $\zeta(5), \ldots, \zeta(11)$ forms is unnecessary for $n \leq 5$ (§8)—apparently the first direct computation of those coefficients.

4. We propose a mechanism (§6): the elimination of $\zeta(2)$ forces the linear form to be the period of a *non-primitive* integral Betti class, and the identity $\zeta(2) = -(2\pi i)^2/24$ prices any integral elimination at a lattice index dividing 24. An elementary lemma makes this precise; the measured constants 12 and 2 then *measure* 2-adic indices of the integral Betti lattice of the underlying rank-3 mixed Tate motive. In the totally symmetric case we exhibit the factor concretely: the excess enters through the doubled Laurent coefficient $\frac{1}{6}$ of the reciprocal-sine kernel, and we prove unconditionally that no prime $p \geq 5$ contributes (§7).

5. All measured denominator slack (cells where the true denominator is *smaller* than predicted) is bounded along every measured direction, including growth series $n \leq 8$; consequently the construction's "worthiness" budget receives no hidden arithmetic subsidy. We also determine $\sup \gamma = 0.86597135\ldots$ over the full 7-dimensional projective parameter cone—attained exactly at the authors' published point, whose symmetric coordinates form an arithmetic progression (§9).

Throughout, "verified" means: exact integers Q_n from the closed binomial double sum of [1]; values of $I(\mathbf{a}\cdot n)$ computed to a certified precision (hundreds of digits) from the 1-dimensional integral of a product of two ${}_3F_2$'s given in [1], evaluated by an exact partial-fraction/polylogarithm method; and $P_n, \hat{P}_n$ recovered by PSLQ over the basis $\{1, \zeta(2)\}$ with residual certification, precision-stability checks, and exact agreement with all rational values printed in [1] (§2).

2 The family and the measurement pipeline

We follow the notation of [1]. For admissible integer parameters $\mathbf{a} = (a_1, \ldots, a_8)$ (the 17 convergence forms nonnegative; we require all 28 forms of the multiset $\mathbf{h}(\mathbf{a})$ nonnegative so that the full symmetry orbit is admissible), the cellular integral decomposes as

$$I(\mathbf{a}) \;=\; 2Q\,(\zeta(5) + 2\zeta(3)\zeta(2)) \;-\; 4\hat{P}\,\zeta(2) \;-\; 2P, \qquad Q \in \mathbb{Z},\; P, \hat{P} \in \mathbb{Q},$$

and the object of interest is $I' = Q\zeta(5) - P$. The group $\mathfrak{G} \cong \Sigma_7$ acts on normalized integrals through the symmetric parameters $s_0; s_1, \ldots, s_7$, permuting the 28-element multiset $\mathbf{h}$, with $I(\mathbf{a})/\prod_{i\in\mathcal{F}} h_i!$ invariant for the 17-element index set $\mathcal{F}$ of [1].

Exact Q. $Q(\mathbf{a}\cdot n)$ is computed exactly from the double binomial sum [1, eq. (sumQ)].

Certified I. $J(\mathbf{p};\mathbf{q}) = I(\mathbf{a})$ reduces [1] to a 1-dimensional integral of a product of two ${}_3F_2$'s. For integer parameters each ${}_3F_2$ coefficient is a proper rational function of the summation index; its exact partial fractions over $\mathbb{Q}$ convert every evaluation into finitely many polylogarithms $\mathrm{Li}_m(z)$, $z \le 0$, with rational weights (analytic continuation built in). Working precision is raised until two runs agree; tanh–sinh quadrature errors are certified against the target precision. On validation, this evaluator agrees with the direct (slow) hypergeometric path to all compared digits (≥ 40) and is $\sim 57\times$ faster on production cells.

Recovery. With ζ-values to matching precision, $x = (2Q\zeta(5) + 4Q\zeta(3)\zeta(2) - I)/2 = P + 2\hat{P}\zeta(2)$, and PSLQ on $(x, 1, \zeta(2))$ returns $P, \hat{P}$ as exact rationals. Certification: the relation residual must vanish to half the working precision; the recovery must be stable under a +80-digit rerun; and the pipeline must reproduce the exact printed values of [1] in the totally symmetric case,

$$Q_1 = 21, \quad P_1 = \tfrac{87}{4}, \quad \hat{P}_1 = \tfrac{101}{4}, \qquad Q_2 = 2989, \quad P_2 = \tfrac{1190161}{384}, \quad \hat{P}_2 = \tfrac{344923}{96},$$

which it does. An incidental end-to-end validation appears in §5: two independent recoveries at distinct parameter points agree with the exact factorial-ratio identity $I'(g\mathbf{a}) = I'(\mathbf{a}) \cdot \prod h_i(\mathbf{a})!/\prod h_i(g\mathbf{a})!$ through a discrepancy of 7^2.

3 The published inclusions are false as stated

Brown–Zudilin state (experimentally) the inclusion $d_{m_1 n} \cdots d_{m_5 n}\, I'(\mathbf{a}\cdot n) \in \mathbb{Z}\zeta(5) + \mathbb{Z}$, where $m_1 \ge \cdots \ge m_5$ are the five largest entries of $\mathbf{h}(\mathbf{a})$, together with a sharpening by the factor $\Phi_n = \prod_{p > \sqrt{m_1 n}} p^{\nu_p}$, and, in the totally symmetric case, the display $Q_n,\ d_n^2 d_{2n}\hat{P}_n,\ d_n^5 P_n \in \mathbb{Z}$.

Already at $n = 2$ in the totally symmetric case, with the paper's own printed values: $d_2^5 P_2 = 2^5 \cdot \frac{1190161}{384} = \frac{1190161}{12} \notin \mathbb{Z}$ (off by $2^2 \cdot 3$), and $d_2^2 d_4 \hat{P}_2 = 48 \cdot \frac{344923}{96} = \frac{344923}{2} \notin \mathbb{Z}$ (off by 2). The numerators are coprime to 6, so the failures are exact. Our recovered values agree with the printed ones, independently. Analogous failures occur at 3 of 19 small interior points in our first systematic sample, always 2-adically. The corrected statements are the subject of the next section.

4 The sharp correction: data and conjecture

Extend Brown–Zudilin's sharpening to every prime (legitimate, since the factorial-ratio identity above is exact p-adically):

$$D_\nu(\mathbf{a}, n) = \frac{\prod_{i=1}^5 d_{m_i n}}{\Phi_n^*}, \qquad \Phi_n^* = \prod_{\text{all } p} p^{\nu_p}, \qquad \nu_p = \max_{g \in \mathfrak{G}} \operatorname{ord}_p \frac{\prod_{i\in\mathcal{F}} (h_i(\mathbf{a})n)!}{\prod_{i\in\mathcal{F}} (h_i(g\mathbf{a})n)!}.$$

Conjecture 4.1 (robust form)**.** *For all admissible* $\mathbf{a}$ *and* $n \ge 1$*, and every prime* p*,*

$$\operatorname{ord}_p \operatorname{den}(P_n) \ \le\ \operatorname{ord}_p D_\nu(\mathbf{a}, n) \ +\ \begin{cases} 2, & p = 2, \\ 1, & p = 3, \\ 1, & p \ge 5, \end{cases}$$

with the ceilings at 2 *and* 3 *attained, the* $p \ge 5$ *term nonzero only at boundary primes* $p = m_1$*, and, in the* ***strong form***, *vanishing for* $n \ge 2$.

We emphasize the epistemic status: both forms are falsifiable by a single future cell, and the boundedness of the $p \geq 5$ term—like every sharpness statement here—rests on finite evidence only. We know of no structural argument excluding infinitely many boundary-prime fluctuations; §5 documents the one observed orbit and the failed attempt to make it recur.

Evidence. Over 200 certified cells (systematic random interior points with entries up to 7, growth series at selected points with $n \leq 8$, and the authors' record point): the 2-adic excess distribution over the first 164 cells is $\{-6{:}2,\ -5{:}1,\ -4{:}4,\ -3{:}9,\ -2{:}9,\ -1{:}20,\ 0{:}33,\ +1{:}40,\ +2{:}46\}$ — ceiling $+2$, attained 46 times, never exceeded; the 3-adic distribution is $\{-4{:}1,\ -3{:}3,\ -2{:}8,\ -1{:}43,\ 0{:}133,\ +1{:}30\}$ — ceiling $+1$, attained, never exceeded. In the totally symmetric case the 2-adic ceiling is attained at every computed n ($n = 1, \dots, 5$ and 8): $\mathrm{den}(P_n) = 4 \cdot 2^{\mathrm{ord}_2 d_n^5} \cdot$ (odd part), and $\mathrm{den}(\hat{P}_n) = 2 \cdot 2^{\mathrm{ord}_2 d_n^2 d_{2n}} \cdot$ (odd part) for $n \leq 5$. The odd parts show only slack (the 3-adic prediction over-pays at most points, which is why the 3-part of the correction is visible only where the 3-adic prediction is tight).

Growth. Along every measured series the excess/slack trajectory is bounded: e.g. the largest observed slack point $\mathbf{a} = (4, 3, 4, 4, 3, 2, 4, 2)$ has 2-adic slack $6, 5, 4, 5, 3$ bits at $n = 1, \dots, 5$; points with attained ceiling at $n = 1, 2$ can relax into slack at $n = 3$. The conjectured bound—not any pointwise formula—is the stable object.

5 A boundary-prime anomaly, and orbit coherence

Exactly one sampled orbit violates the $p \geq 5$ clause: at $\mathbf{a} = (6, 2, 6, 1, 6, 4, 5, 5)$, $n = 1$ (where $m_1 = 7$ is prime and $\nu_7 = 1$), $\mathrm{den}(P_1) = 2^7 \cdot 3^4 \cdot 5^2 \cdot 7$ exceeds the ν-sharpened prediction by one factor of 7. Since the factorial-ratio identity is exact, this *predicts* that the maximizing orbit partner $\mathbf{a}' = (6, 2, 5, 2, 4, 6, 6, 5)$ must violate even the raw (ν-free) law; direct audit confirms $\mathrm{den}(P_1(\mathbf{a}')) = 2^6 \cdot 3^4 \cdot 5^2 \cdot 7^2$ against a d-product paying a single 7. Excesses are thus *orbit-coherent*: the correct invariant is an orbit-level anomaly divisor.

In our samples the effect is rare and non-persistent: a targeted scan of 40 fresh points satisfying the same trigger condition ($m_1 \in \{5, 7, 11, 13\}$ prime, $\nu_{m_1} \geq 1$) produced *no* further instance, and the anomalous orbit itself relaxes to slack (-2 at $p = 7$) at $n = 2$, at both partners coherently. We stress that this falsifies the naive "$\{2, 3\}$ only" clause outright, and that finite sampling cannot exclude infinitely many such violations across the cone or along n; the robust form of Conjecture 4.1 prices them at one unit at the boundary prime, and even that bound is an empirical reading, not a theorem. Characterizing the boundary-prime effect exactly is open, and we regard it as the sharpest known probe of what the $\mathfrak{G}$-symmetrized inclusion machinery misses.

6 The mechanism

Lemma 6.1 (elimination cost)**.** *Let e_0, e_2 span a 2-dimensional $\mathbb{Q}$-vector space and let L be a lattice in the dual with basis γ_1, γ_2 pairing as $\langle \gamma_1, (e_0, e_2) \rangle = (1, \zeta(2))$, $\langle \gamma_2, (e_0, e_2) \rangle = (0, (2\pi i)^2)$. Every nonzero $\delta \in L$ with $\langle \delta, e_2 \rangle = 0$ is an integer multiple of $\delta_0 = 24\gamma_1 + \gamma_2$; in particular $\langle \delta, e_0 \rangle \in 24\mathbb{Z}$.*

Proof. $a\zeta(2) + b(2\pi i)^2 = 0$ and $\zeta(2)/(2\pi i)^2 = -\frac{1}{24}$ force $b = a/24$, so $24 \mid a$. □

If the true integral lattice is finer, with $\gamma_2/2^k$ integral, the cost drops to $24/2^k$. The linear form I' arises exactly by such an elimination: $\mathrm{Ext}^1_{MT(\mathbb{Z})_\mathbb{Q}}(\mathbb{Q}(0), \mathbb{Q}(2)) = 0$ [5] splits the weight-4 part rationally, and the identity $\zeta^{\mathfrak{m}}(2) = -(\mathbb{L}^{\mathfrak{m}})^2/24$ is available motivically [6, §2.2]. Under two explicitly stated hypotheses—(H1) the integrand's de Rham coefficients are D_ν-controlled (the integral form of Brown–Zudilin's own sketched Barnes analysis), and (H2) a normalization of the integral Betti lattice—one obtains $24\, D_\nu\, I' \in \mathbb{Z}\zeta(5) + \mathbb{Z}$, with the constant improving by the lattice index. The measured sharp constants then *measure* that index: $12 = 24/2$ for P (equivalently, cost 2 on the natural coefficient $2P$), and 2 for $\hat{P}$. To our knowledge these are the first experimentally measured integral-lattice invariants of periods of $\overline{\mathcal{M}}_{0,8}$; the literature contains the $\mathbb{Q}$-theory [8, 5, 7] and an empirical $\{2,3\}$-clearing phenomenon for motivic period expansions [9], but—so far as we can determine—no integral-lattice synthesis. The support $\{2,3\}$ matches the torsion $K_3(\mathbb{Z}) \cong \mathbb{Z}/48$ [10] of the relevant splitting obstruction; we flag this as a consistency, not a derivation.

7 The totally symmetric case: concrete derivation and a support theorem

In the totally symmetric case $\mathbf{a} = (1, \dots, 1)$ the Barnes representation reduces to the reciprocal-sine kernel with rational part $r(s)r(t)C(s,t)$, $r(s) = \prod_{i=1}^{n}(s+i)/\prod_{j=n+1}^{2n+1}(s+j)^2$, whose partial-fraction data is fully explicit: $B_j = (-1)^n\binom{n+a}{a}\binom{n}{a}^2/n!$ and $A_j = B_j\,(3H_a - H_{n+a} - 2H_{n-a})$ with $a = j-n-1$ (verified exactly for $n \le 6$). Two consequences, proved in the accompanying notes:

Theorem 7.1 (support; proof in the companion notes, under independent audit)**.** *In the totally symmetric case every prime dividing* $\mathrm{den}(P_n)$ *is at most* $2n+1$*, and no prime* $p \ge 5$ *exceeds the* d_n^5 *budget.*

Status. The proof is written in the companion computational notes and its instances are machine-verified for $n \le 6$; it will be reproduced in this paper only after a full independent audit, and until then the reader should weight it accordingly.

Moreover the $\{2,3\}$-excess enters *only* through the Laurent coefficients $1, \frac{1}{6}, \frac{1}{90}, \dots$ of the sine kernel: the constant term P_n picks up exactly one doubled factor of the $\zeta(2)$-coefficient $\frac{1}{6}$, i.e. the factor $12 = 2 \cdot 6$—the concrete avatar of Lemma 6.1—while $\frac{1}{90}$ (which would inject a 5) provably does not contribute, matching Theorem 7.1 and the data. The remaining gap to a full proof of $12\, d_n^5 P_n \in \mathbb{Z}$ is a classical-type d_n^5-collapse bookkeeping for the structured sums above; it is verified exactly for $n \le 5$.

8 Cross-family controls

The same measurement standard applied to the classical corpus shows its correction factors are uniformly *removable*, in contrast to §4: (i) Apéry $\zeta(3)$: here removability is a *theorem*, not merely our finite check: Rhin–Viola [4, Thm. 2.1] prove $d_n^3 a_n \in \mathbb{Z}$ exactly (the genuine 2 rides on the $\zeta(3)$-coefficient parity, not the denominator); our $n \le 50$ verification is a sanity check of their statement. (ii) $\zeta(4)$: $d_n^4 v_n \in \mathbb{Z}$ for $n \le 5$ in Zudilin's derived series (the factor 2 retained in [3, Théorème 3] is slack there). (iii) Zudilin's asymmetric $\zeta(5), \dots, \zeta(11)$ forms: computing the five coefficients $z_{j,n}$ exactly for $n \le 5$ (apparently for the first time), the leading factor 2 retained even in the *conjectured* law [3, eq. (17.1)] is unnecessary, with ~ 30 bits of 2-adic margin; the well-poised parity cancellation and the proven inclusion of [3] serve as engine validations. For

(ii) and (iii) removability is genuinely open (Krattenthaler–Rivoal prove their inclusions only "à un facteur 2 près" and could not remove it); our checks are finite verifications, and the structural reason to expect removability there is identifiable but heuristic: the factor 2 in the proofs arises from termwise 2-adic over-counting at half-integer shifts, which the full sums have no reason to respect. What the controls establish is a contrast, not a theorem: in every classical family examined the correction is unneeded in all computed cases, while in the Brown–Zudilin family—the only construction here that performs a $\zeta(2)$-elimination—it is provably needed (the printed values of §3 already require it).

9 The worthiness supremum, and what the arithmetic cannot do

For completeness we record the analytic side. Implementing the worthiness exponent $\gamma(\mathbf{a})$ of [1] (validated to 8 decimals against all three published values, including every printed intermediate), a global search of the 7-dimensional projective cone —steepest ascent from the published point on integer and half-integer lattices, 120 random multi-starts, 60 basin hops, and an exhaustive scan of all 31,710 projectively distinct arithmetic-progression points to height 160—finds the supremum

$$\sup_{\mathbf{a}} \gamma \;=\; 0.86597135\ldots,$$

attained exactly at the published point $\mathbf{a} = (8, 16, 10, 15, 12, 16, 18, 13)$, whose symmetric coordinates $(20.5; 3.5, 4.5, \ldots, 9.5)$ form an arithmetic progression. Combined with the bounded-slack measurements of §4, the construction's arithmetic is now understood to bounded factors, and none of it moves the exponential budget: within this family the barrier to $\gamma > 1$ is structural.

10 Limitations and outlook

All sharpness and slack statements are finite verifications under the certification protocol of §2; Conjecture 4.1 is falsifiable by a single future cell. The mechanism of §6 is proved only in the elementary Lemma 6.1 plus the conditional reduction; hypothesis (H2) is one finite lattice computation (torsion-freeness of $H^*(\overline{\mathcal{M}}_{0,n}, \mathbb{Z})$ [11] should make it tractable), and (H1) is the integral form of the Barnes bookkeeping sketched in [1]. The measurement pipeline itself is portable, and we suggest its use as an *arithmetic screen* for any future candidate constructions: the true denominators of a proposed family can now be measured in hours rather than conjectured. In a companion development, Lemma 4 of [2] (the asymptotic core of the elementary proof that one of $\zeta(5), \ldots, \zeta(25)$ is irrational) has been fully formalized in Lean 4/Mathlib, with a complete formalization of the theorem in progress; details will appear separately.

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